

C. Flexible String

time limit per test: 2 seconds
memory limit per test: 256 megabytes

You have a string a and a string b . Both of the strings have length n . There are **at most 10 different characters** in the string a . You also have a set Q . Initially, the set Q is empty. You can apply the following operation on the string a any number of times:

- Choose an index i ($1 \leq i \leq n$) and a lowercase English letter c . Add a_i to the set Q and then replace a_i with c .

For example, Let the string a be "abecca". We can do the following operations:

- In the first operation, if you choose $i = 3$ and $c = \mathbf{x}$, the character $a_3 = \mathbf{e}$ will be added to the set Q . So, the set Q will be $\{\mathbf{e}\}$, and the string a will be "abxcca".
- In the second operation, if you choose $i = 6$ and $c = \mathbf{s}$, the character $a_6 = \mathbf{a}$ will be added to the set Q . So, the set Q will be $\{\mathbf{e}, \mathbf{a}\}$, and the string a will be "abxccs".

You can apply any number of operations on a , but in the end, the set Q should contain **at most k different characters**. Under this constraint, you have to maximize the number of integer pairs (l, r) ($1 \leq l \leq r \leq n$) such that $a[l, r] = b[l, r]$. Here, $s[l, r]$ means the substring of string s starting at index l (inclusively) and ending at index r (inclusively).

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line contains two integers n and k ($1 \leq n \leq 10^5$, $0 \leq k \leq 10$) — the length of the two strings and the limit on different characters in the set Q .

The second line contains the string a of length n . There is **at most 10 different characters** in the string a .

The last line contains the string b of length n .

Both of the strings a and b contain only lowercase English letters. The sum of n over all test cases doesn't exceed 10^5 .

Output

For each test case, print a single integer in a line, the maximum number of pairs (l, r) satisfying the constraints.

Example

input	output
6	6
3 1	3
abc	6
abd	6
3 0	6
abc	6
abd	11
3 1	
xbb	
xcd	
4 1	
abcd	
axcb	
3 10	
abc	
abd	
10 3	
lkwhbahuqa	
qoiujoncj b	

Note

In the first case, we can select index $i = 3$ and replace it with character $c = \mathbf{d}$. All possible pairs (l, r) will be valid.

In the second case, we can't perform any operation. The 3 valid pairs (l, r) are:

1. $a[1, 1] = b[1, 1] = \mathbf{a}$,
2. $a[1, 2] = b[1, 2] = \mathbf{ab}$,
3. $a[2, 2] = b[2, 2] = \mathbf{b}$.

In the third case, we can choose index 2 and index 3 and replace them with the characters $\mathbf{c}$ and $\mathbf{d}$ respectively. The final set Q will be $\{\mathbf{b}\}$ having size 1 that satisfies the value of k . All possible pairs (l, r) will be valid.